Changes in Adolescent Alcohol Use: Longitudinal Associations With Peer Drinking and Parental Alcoholism

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April 6, 2026

Introduction

Alcohol use often begins in adolescence, a period during which social and family influences can shape behaviour. The objective of this study was to determine whether individual trajectories of adolescent alcohol use differ according to a history of parental alcoholism and early peer alcohol use. Data were collected on the longitudinal drinking habits of 82 adolescents, totaling to 246 observations over three years. Participants completed behavioural questionnaires annually at ages 14, 15, and 16. The outcome of interest was alcohol consumption in the previous year, measured as a composite score ranging from 0 (low use) to approximately 5.3 (high use), calculated as the square root of the sum of responses across a four-item instrument. Parental alcoholism (COA status) was assessed as a binary variable based on self-reporting. The questionnaires also assessed peer alcohol use, which was measured using a two-item instrument and similarly transformed, ranging from 0 (low use) to approximately 3.2 (high use). COA status, peer alcohol use, and gender were only measured at the baseline age 14.

Methods

Descriptive Analyses

The initial analysis assessed the sample and data completeness. Baseline records were defined as the first observation for each participant at age 14. Subject-level characteristics of peer alcohol use, gender, and COA status were summarized and visualized for any covariate trends. Alcohol use was summarized using the mean, standard deviation, median, and range at ages 14, 15, and 16 since age is the study's time component. Baseline peer alcohol use was summarized using the same descriptive statistics, while gender and COA status were summarized using counts and percentages for each factor level. Missing data were examined by age for all covariates. Exploratory data analysis included boxplots, individual trajectory plots, and line plots to assess distributions, longitudinal patterns, and overall trends in alcohol use.

Covariance Structure Selection

Since the data consisted of repeated measurements collected from the same adolescents over time, linear mixed-effects models were used to account for within-subject correlation and to evaluate individual trajectories of alcohol use. Random effects accounted for between-subject differences in baseline alcohol use and trajectories over time. A covariance structure was selected

using restricted maximum likelihood (REML). Age, COA status, peer alcohol use, and gender were included as the initial predictors of interest. To determine the optimal random-effects and residual error structure, five covariance structures were compared: 1) a random intercept model with independent errors, 2) a random intercept and random slope (for age) model, 3) a random intercept model with first-order autoregressive (AR(1)) errors, 4) a random intercept model with heteroskedastic independent errors by age, and 5) a random intercept model with heteroskedastic independent errors by gender. Model fit was assessed using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). The model with the lowest AIC and BIC was selected for further analysis.

Mean Model Selection

After selecting the covariance structure, fixed-effects selection was carried out using maximum likelihood (ML). The following nested structure was compared with likelihood ratio tests (LRT): 1) a model including only main effects for age, COA status, and peer alcohol use, 2) a model including all two-way interactions among age, COA status, and peer alcohol use, and 3) a model including both two-way and three-way interactions. The same mean structures with gender as an added main effect were also considered. Final model selection was informed by the LRT, AIC, and BIC results. Although more complex models may better capture trends in alcohol use, parsimony and interpretability were also considered when selecting the final model.

Model Diagnostics

For the selected model, diagnostics were performed using residual plots, including histograms of raw and standardized residuals, normal quantile-quantile (Q-Q) plots, residuals versus fitted value plots, and lagged residual plots. Normality was assessed using QQ-plots and histograms of both residuals and random effects, where deviations from the diagonal line or skewness indicated potential violations. Homoscedasticity was assessed using residuals versus fitted plots, where trends or fanning suggest departures from model assumptions. Remaining within-subject correlation was evaluated using lagged residual plots to assess whether the correlation structure adequately accounted for dependence.

If diagnostic plots from the selected model indicated potential violations of model assumptions, an alternative structure was considered and selected based on improved diagnostics. Specifically, the second-best covariance structure identified during initial model comparison was considered. Under this structure, the same fixed-effects selection procedure was repeated using maximum likelihood, including comparisons of models with main effects, two-way interactions, and three-way interactions. Diagnostic assessments were then repeated to evaluate whether model assumptions were better satisfied under this specification.

Final Model Predictions

Predicted alcohol use trajectories from the final model were generated for interpretation using subject-specific (conditional) and population-averaged (marginal) estimates. Marginal predictions fixed peer alcohol use at its sample mean, while additional predictions were evaluated at one standard deviation below and above the mean to represent low and high peer use. Predictions were computed across ages 14 to 16 for those with and without parental alcoholism.

Statistical Software

All analyses were conducted in R (version 4.5.1) using the packages `tidyverse`, `nlme`, `lme4`, `car`, `AICcmodavg`, `patchwork`, `tibble`, `knitr`, `kableExtra`, `here`, and `tinytex`. These were used for data manipulation, mixed-effects modeling, model comparison, visualization, and report generation.

Results

Descriptive Analyses

Table 1 summarizes the primary outcome of alcohol use by age and baseline covariates, including peer alcohol use, gender and COA status. On average, mean alcohol use increased with age, as 15 and 16-year old subjects reported higher and more variable alcohol use compared to 14 year olds. The highest alcohol use score (3.61) was observed at age 15. A substantial proportion of adolescents reported no alcohol use, particularly at age 14 (median = 0), indicating a concentration of zero values at younger ages. Alcohol use amongst 16 year olds was positively skewed, and showed a narrower range compared to 15 year olds. Peer alcohol use had a relatively low mean of 1.02 (SD = 0.73). Males and females were similarly represented in the sample (51.2% vs. 48.8%), while subjects without an alcoholic parent were slightly overrepresented (54.9% vs. 45.1%). All 82 participants completed the study, with no missing data for any aforementioned covariates.

Table 1: Descriptive Statistics of Study Variables

Characteristic	Age 14	Age 15	Age 16
Alcohol Use			
Mean (SD)	0.63 (0.94)	0.96 (1.05)	1.17 (1.14)
Median	0.00	1.00	1.00
Range	0.00–3.32	0.00–3.61	0.00–3.46
Baseline Characteristics			
Peer Alcohol Use			
Mean (SD)	1.02 (0.73)		
Median	0.89		
Range	0.00–2.53		
Gender			
Female	40 (48.8%)		
Male	42 (51.2%)		
COA Status			
No	45 (54.9%)		
Yes	37 (45.1%)		

* COA status indicates parental alcoholism (child of an alcoholic).

** Baseline characteristics were assessed at age 14 and assumed to remain unchanged at ages 15 and 16.

Exploratory Analyses

Exploratory analyses revealed consistent differences in alcohol use by COA status and peer alcohol use. Across all ages, adolescents with an alcoholic parent exhibited higher alcohol use and greater variability compared to those without (Figure 1). In addition, alcohol use increased

with age in both groups, and the magnitude of differences between COA groups appeared to become more pronounced at older ages, suggesting a potential interaction between age and COA status.

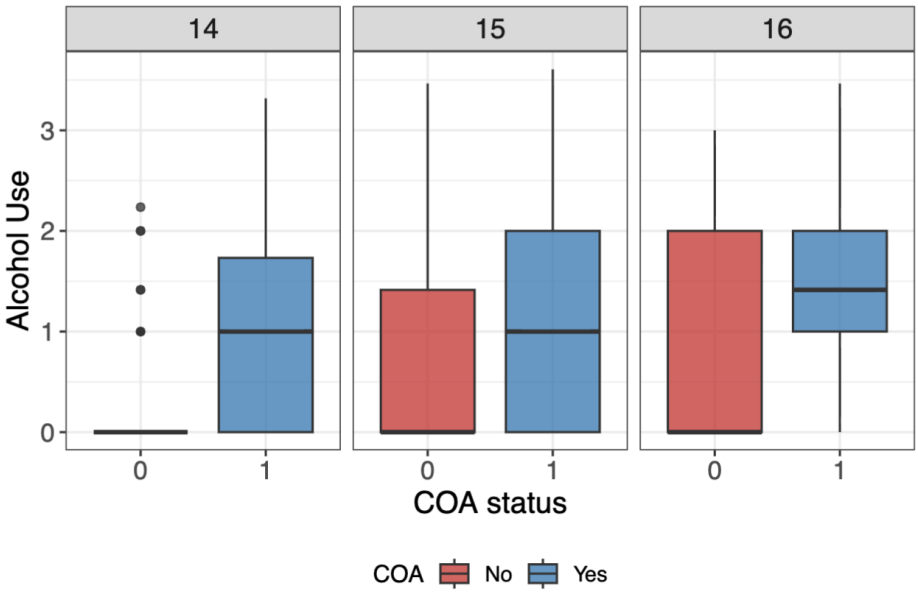

Figure 1: Distribution of alcohol use by parental alcoholism (COA status), stratified by age (14–16). Boxplots summarize the median, interquartile range, and variability within each group.

A positive association between peer alcohol use and individual alcohol use was observed at each age (Figure 2). This relationship was consistently positive and stronger among adolescents with an alcoholic parent. Furthermore, the strength of this association appeared to differ across age, suggesting a potential interaction between age, peer alcohol use, and COA status.

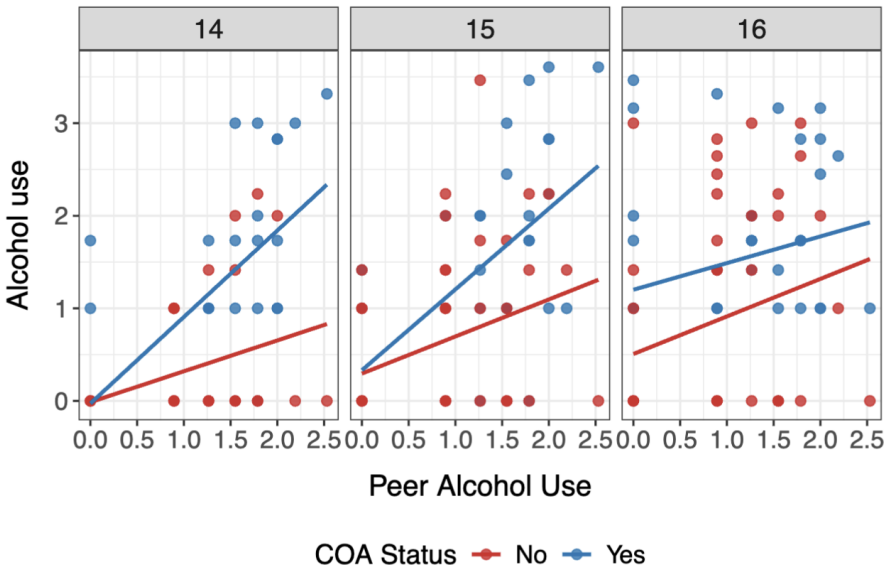

Figure 2: Association between peer alcohol use and individual alcohol use, stratified by age (14-16 years) and parental alcoholism (COA status). Points represent individual observations, and lines indicate fitted linear trends.

Longitudinal trajectories of alcohol use (Figure 3) showed that adolescents with an alcoholic parent consistently exhibited higher mean alcohol use than adolescents with a non-alcoholic parent across ages 14 to 16. While alcohol use generally increased over time in both groups, there was substantial between-participant variation in the baseline alcohol use and the trajectory of alcohol use by age (rates of change). Individual trajectories varied considerably, with some adolescents showing little change and others exhibiting sharper increases or fluctuations over time. These patterns motivated the use of subject-specific random effects such as random intercepts and slopes to account for heterogeneity in baseline alcohol use and individual trajectories. Although the trajectories of alcohol use appear to change direction at age 15 for some individuals, a linear trend from age 14-16 was assumed for model parsimony.

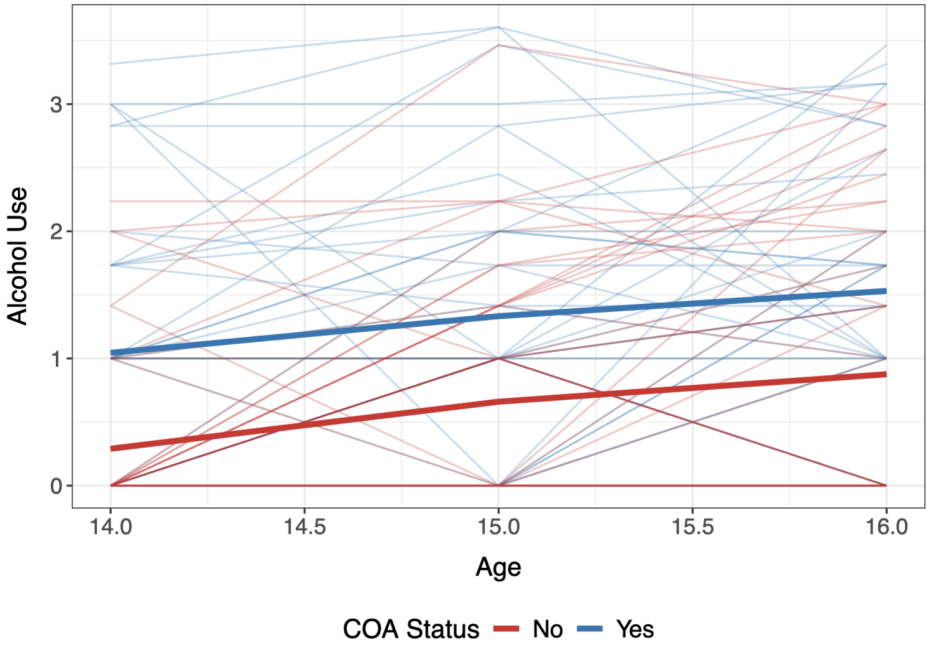

Figure 3: Individual trajectories of alcohol use from ages 14 to 16, stratified by parental alcoholism (COA status). Thin lines represent individual participants, and thick lines indicate average trends within each group.

Covariance Structure Selection

Candidate covariance structures were compared using AIC and BIC in Table 2. The random intercept and slope model with independent errors provided the best overall fit, with the lowest AIC (626.2) and BIC (671.3), followed by the random intercept model with AR(1) errors (AIC = 634.7, BIC = 676.4). In other words, these models used the least parameters necessary to fit the data and prevent overfitting. Initial diagnostics (Figure A1) indicated violations of model assumptions for the random intercept and slope model, particularly with respect to residual normality. In contrast, the random intercept with AR(1) errors performed better in residual checks (Figure A2) and was therefore selected as the final covariance structure. The AR(1) structure was also appropriate given the equally spaced longitudinal measurements, as it accounts for within-subject correlation over time.

Table 2: Comparison of covariance structures

Covariance model	AIC	BIC
Random intercepts + independent errors	640.750	678.899
Random intercepts and slopes + independent errors	626.169	671.254
Random intercepts + AR(1) errors	634.744	676.360
Random intercepts + heteroskedastic independent errors by age	634.656	679.741
Random intercepts + heteroskedastic independent errors by gender	641.983	683.600

Fixed Effects Selection

Under the random intercept model with AR(1) errors, fixed effects selection involved comparing models of increasing complexity, from main effects to interaction terms (Table 3). The inclusion of two-way interactions did not significantly improve model fit compared to the main effects model (p=0.164). However, the three-way interaction model (age $\times$ peer alcohol use $\times$ COA) performed significantly better than the two-way interaction model ($p = 0.032$), supporting the presence of higher-order interactions.

When gender was included, a similar pattern was observed, with the three-way interaction model showing improved fit compared to the two-way interaction model (p=0.033). Although the model including gender had a slightly lower AIC than the model without gender (612.6 vs 614.0), it was penalized more heavily by BIC (654.6 vs 652.6). Since the association between gender and alcohol use was not part of the original research question, and did not substantially improve model fit, the three-way interaction model without gender was selected for parsimony.

Table 3: Fixed-effects Structure Selection under AR(1) Random Intercept Model

Model	Test	df	logLik	L.Ratio	p-value	AIC	BIC
1. Main effects		7	-300.888	NA	NA	615.777	640.314
2. Two-way interactions	1 vs 2	10	-298.333	5.110	0.164	616.667	651.720
3. Three-way interactions	2 vs 3	11	-296.021	4.624	0.032	614.043	652.602
4. Main effects (with gender)		8	-299.052	NA	NA	614.103	642.146
5. Two-way interactions (with gender)	4 vs 5	11	-296.569	4.965	0.174	615.139	653.697
6. Three-way interactions (with gender)	5 vs 6	12	-294.286	4.567	0.033	612.572	654.636

Final Model Results

The final selected model fit an intercept and fixed effects for age, COA status, peer alcohol use, along with all corresponding two-way and three-way interaction terms (Table 4). The interaction terms were between age and COA status, age and peer alcohol use, peer alcohol use and COA status, and all three aforementioned covariates. The random intercept allowed for individual differences in baseline alcohol use. The significance and magnitude of the three-way interaction are central to addressing the study objective of whether alcohol use trajectories vary jointly by COA status and peer alcohol use.

Table 4: Estimated Coefficients from Final 3-way Interaction Model

Variable	Estimate	SE	95% CI	p-value
Constant	-3.645	2.056	(-7.674, 0.385)	0.078
Age	0.260	0.137	(-0.007, 0.528)	0.059
COA	-5.041	3.176	(-11.267, 1.185)	0.117
Peer alcohol use	-0.164	1.817	(-3.725, 3.398)	0.928
Age × COA	0.354	0.211	(-0.059, 0.768)	0.095
Age × peer alcohol use	0.036	0.121	(-0.201, 0.273)	0.766
COA × peer alcohol use	5.682	2.527	(0.73, 10.634)	0.027
Age × COA × peer alcohol use	-0.359	0.168	(-0.688, -0.03)	0.034

The interactions COA × peer alcohol use (p=0.027) and age × COA × peer alcohol use (p=0.034) were statistically significant ($p<0.05$). These effects captured the influence of peer alcohol use on adolescent drinking changes by COA status as a sole effect modifier and jointly with age. The estimated two-way interaction (5.682) indicated that the relationship between peer alcohol use and subject alcohol use was stronger among COA adolescents, which refers to adolescents with an alcoholic parent. In addition, the negative three-way interaction (-0.359) suggested that alcohol use trajectories varied jointly by age, COA status, and peer alcohol use. The strength of the association between peer and individual alcohol use among COA adolescents decreased slightly with age. Together, these findings provided evidence that adolescent alcohol use trajectories differed by parental alcoholism and peer influence, and that these effects evolved over time.

Predicted Trajectories

The predicted trajectories of the final model are shown in Figure 4. Differences in slope across COA status and peer alcohol use levels reflected the significant three-way interaction. Among non-COA adolescents, or those without an alcoholic parent, trajectories were similar between low and high peer alcohol use, indicating that peer influence had a relatively stable effect over time. In contrast, among COA adolescents, trajectories vared substantially by peer alcohol use. Specifically, those with low peer alcohol use exhibited the steepest increases over time, while those with high peer alcohol use showed flatter trajectories. This pattern suggested that the effect of peer influence on alcohol use changed with age in this group.

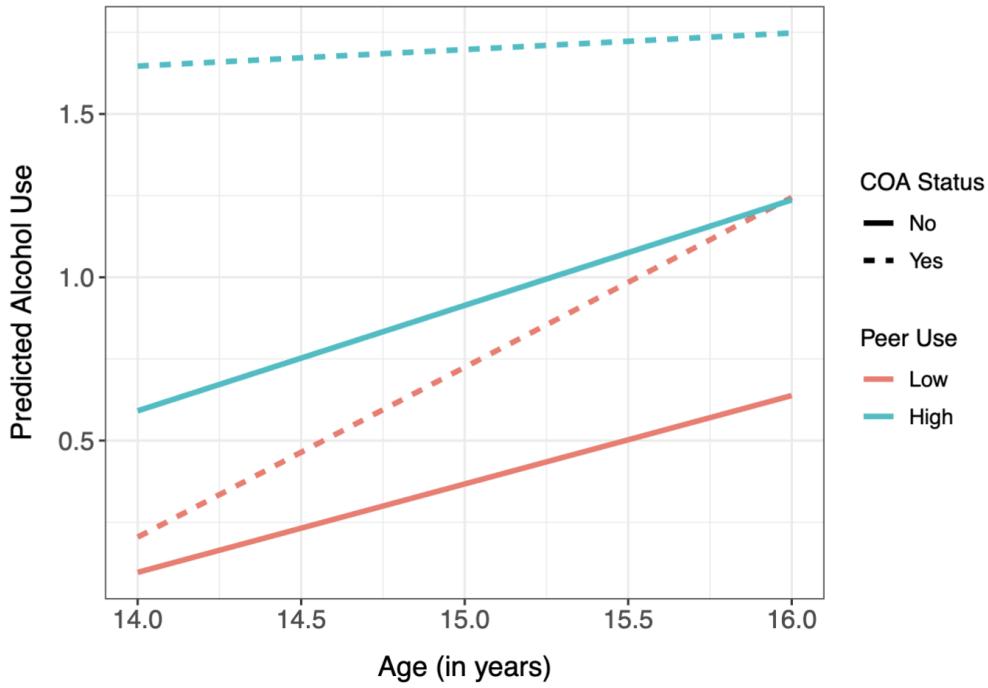

Figure 4: Predicted alcohol use trajectories from the final AR(1) model across ages 14-16, stratified by COA status and peer alcohol use. Peer use was evaluated at one standard deviation below (low = 0.29) and above (high = 1.74) the sample mean (1.02).

These patterns were consistent with the model estimates in Table 4. Among non-COA adolescents with no peer alcohol use, alcohol use increased by 0.260 points per year. Holding peer alcohol use at zero, COA adolescents exhibited an additional increase of 0.354 points per year compared to non-COA adolescents, resulting in a steeper overall trajectory (0.614 points per year). Although these yearly changes were subtle relative to the overall range of alcohol use (0 to 5.3), they reflected meaningful differences in growth over time. In addition, the association between alcohol use and peer drinking varied significantly by COA status. When age was fixed, a one-unit increase in peer alcohol use score was associated with a greater increase in alcohol use among COA adolescents compared to non-COA adolescents.

Marginal mean trajectories were computed for adolescents with and without an alcoholic parent by fixing peer alcohol use at its sample mean ($\text{peer}_0 = 1.02$). The expected alcohol use trajectory for non-COA adolescents was:

$$E(Y_{ij} \mid \text{côa} = 0,\ \text{pêer} = \text{peer}_0) = -3.645 - 0.164\,\text{peer}_0 + \text{age}_{ij}\,(0.260 + 0.036\,\text{peer}_0)$$

When COA status was present, the marginal model was:

$$E(Y_{ij} \mid \text{côa} = 1,\ \text{pêer} = \text{peer}_0) = -8.686 + 5.518\,\text{peer}_0 + \text{age}_{ij}\,(0.874 - 0.323\,\text{peer}_0)$$

The predicted trajectories (Figure 5) show that when peer alcohol use was held constant, differences in trajectory steepness between COA and non-COA adolescents were minimal. However, COA adolescents exhibited higher average alcohol use across all ages from 14 to 16.

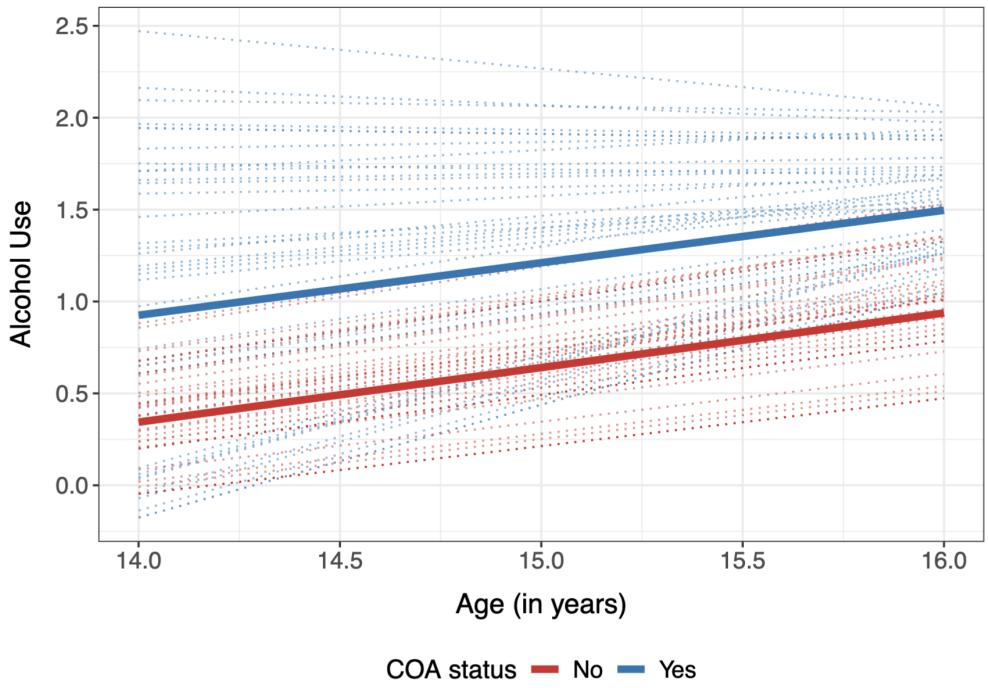

Figure 5: Predicted alcohol use trajectories from the final model across ages 14-16, stratified by parental alcoholism (COA status). Dashed lines represent subject-specific (conditional) predicted trajectories, while solid lines indicate population-averaged (marginal) predicted means evaluated at the average level of peer alcohol use.

Discussion

This study investigated whether personal trajectories of alcohol consumption in adolescence differed by parental alcoholism and early peer alcohol use. Using a linear mixed-effects model, adolescent alcohol consumption was found to vary by age, peer drinking and COA status. COA adolescents with high peer alcohol use exhibited the highest mean alcohol use at each age, but the flattest trajectory over time. One possible explanation is that earlier exposure to alcohol and normalization of drinking in the household may predispose these adolescents to drinking earlier, resulting in elevated baseline levels and smaller subsequent increases with age. In a low peer alcohol environment, COA adolescents showed rapid increases in alcohol use trajectory over time, suggesting greater sensitivity to changes in peer context. Overall, alcohol use increased with age, which is consistent with increasing social acceptance and accessibility of alcohol during adolescence. The findings suggest that drinking habits of COA adolescents are especially sensitive to peer behaviour, highlighting how familial and social environments shape alcohol use trajectories. This interpretation is supported by prior research linking parental substance abuse, including alcoholism, to mental health and substance use issues in teenagers (Jääskeläinen et al., 2016). From a public health perspective, interventions targeting COA adolescents, particularly those in low peer alcohol use environments, may help prevent the largest increases in alcohol use during this developmental period. Non-COA adolescents were also expected to increase alcohol use with age, but their trajectories appeared less influenced by peer context.

This analysis indicated that aging alone did not account for trends in adolescent drinking. Individual trajectories of adolescent alcohol use differed jointly by COA status and early peer alcohol use. These results align with prior research, which has found a combined influence of

peer and parental drinking on adolescent drinking (Olson & Crosnoe, 2018). Future studies should consider alcohol use trajectories over the entire duration of adolescence, beyond ages 14-16. Collecting data on peer alcohol use at each age, as well as other potential predictors such as socioeconomic status, could further inform alcohol reduction programs.

This analysis had some limitations. First, the use of self-report questionnaires for data collection introduced potential reporting bias. Adolescents, in particular, have been found to over- and under-report their substance use (Williams & Nowatzki, 2005). COA adolescents may have been over-represented in this sample, as recent studies estimate only 25% of children have a parent with a substance use disorder (McCabe et. al., 2023). In addition, a key predictor of interest, peer alcohol use, was only measured at baseline. In reality, peer alcohol use likely varies alongside individual alcohol use during adolescence, but such longitudinal data were not available. The data also lacked regional and temporal context. Legal drinking ages and attitudes toward alcohol differ across settings, limiting the generalizability of these findings. From a modeling perspective, trajectories were estimated using a limited number of time points, and a linear trend was assumed. The sample size and number of measurement occasions were relatively small and could be expanded to improve statistical power. It was also assumed that the composite, transformed measures of personal and peer alcohol use accurately reflected alcohol consumption. Finally, model interpretation was somewhat limited since alcohol use score was derived from a four-item composite instrument.

Overall, these findings suggest that adolescent alcohol use varies by age, parental alcoholism, and peer influence, and that these factors interact to shape individual trajectories. This highlights the importance of considering both familial and social contexts in future research and intervention strategies.

References

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Appendix A

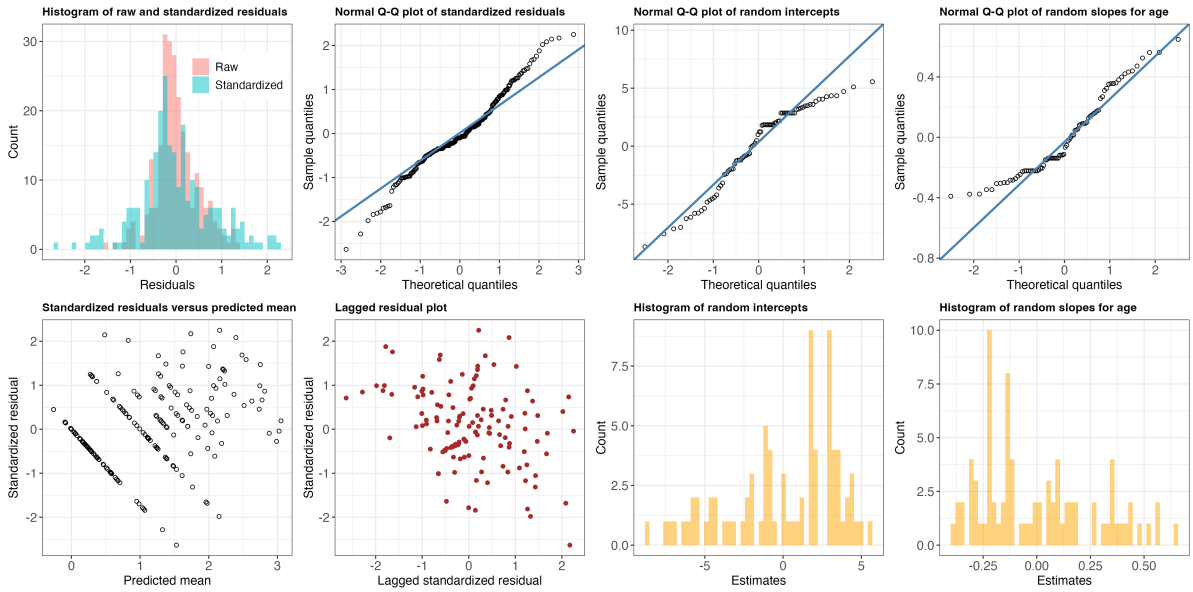

Figure A1: Diagnostic plots for the random-intercept and random-slope model. Fixed effects included age, parental alcoholism status (COA), peer alcohol use, and their two-way and three-way interaction terms. Panels show histograms and normal Q-Q plots of raw and standardized residuals, residuals versus predicted values, lagged residuals, and the estimated random intercepts and random slopes for age. These plots were used to assess normality, homoscedasticity, and remaining within-subject correlation.

The QQ plots showed deviation from the diagonal lines, indicating a violation of normality for the residuals, random intercepts, and random slopes. The histograms of random effects were also not centered at 0, violating the assumption of a zero mean. From this model, fixed effect estimates would still be unbiased, but inference could be unreliable.

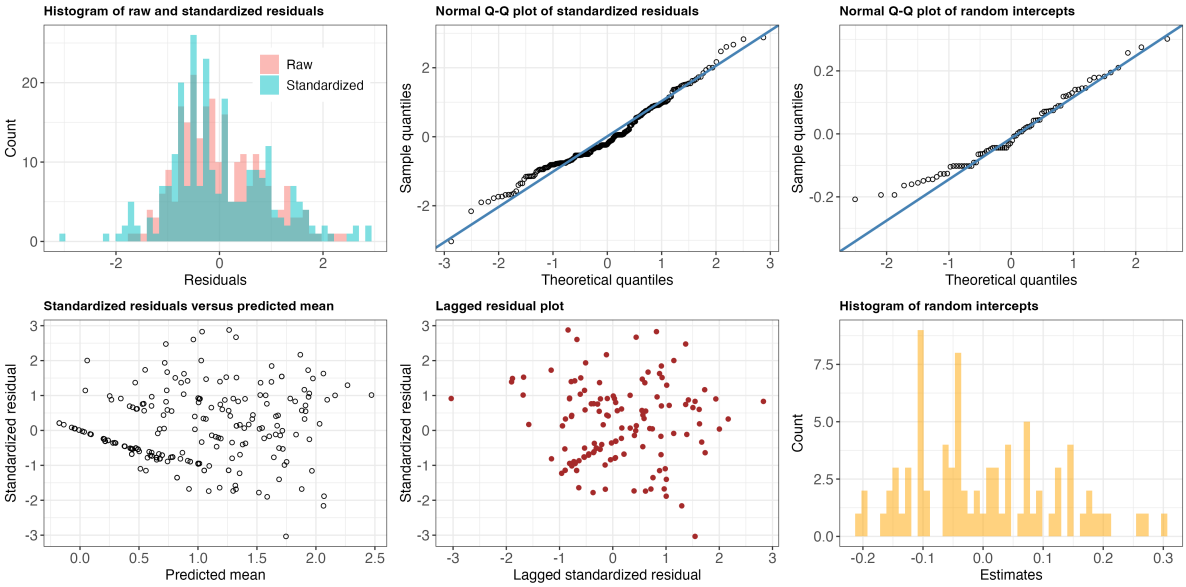

Figure A2: Diagnostic plots for the random-intercept model with an AR(1) correlation structure. Fixed effects included age, parental alcoholism status (COA), peer alcohol use, and their two-way and three-way interaction terms. Panels include histograms and normal Q-Q plots of residuals, residuals versus predicted values, lagged residuals, and the estimated random intercepts. Overall, residual diagnostics suggest approximate normality and improved model fit compared to the random intercept and slope model, although some deviation remains in the random intercept distribution.

Compared to the random intercept and slope model, this model performed much better on residual checks. Model residuals were approximately normal, although the random intercepts still showed some deviation. For the most trustworthy inference, this model was selected.